

Physics-Guided Latent Relighting with Geometry-Aware StyLitGAN

Machine Vision Project Report

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[GitHub:Geometry_and_Physics_Aware_StyLitGAN](#)

Abstract

Image relighting with StyleGAN-like generators is attractive because it avoids explicit three-dimensional reconstruction, but purely latent methods such as StyLitGAN often ignore geometry and physics, so highlights and shading drift in ways no plausible lighting setup could explain. This project investigates whether approximate geometry and a physically based image-formation model can regularise latent relighting while preserving the flexibility of StyleGAN2. A first prototype uses StyLitGAN with Sobel-based normals, log-domain intrinsic optimisation and a Lambertian renderer, which was used almost as a proof of concept. A second, final system replaces these components with MiDaS depth and perspective-correct normals, a neural intrinsic decomposer, spherical harmonics lighting, a Cook–Torrance BRDF and multi-scale, light-conditioned latent directions trained with perceptual losses. Experiments on StyleGAN2 bedroom scenes show that this final system produces stronger and more coherent relighting, better albedo preservation and substantially higher training efficiency than both the StyLitGAN baseline and the initial prototype.

1 Introduction

Image relighting is a central problem in vision and graphics, with applications in data augmentation, robustness testing, virtual staging and interactive VR/AR editing. Many of these scenarios assume fixed scene geometry and materials under varying illumination, but typical datasets expose only a few lighting conditions and classical pipelines require explicit geometry and reflectance estimation, which is brittle and expensive for cluttered indoor scenes.

StyleGAN-based generators offer an alternative. Methods such as StyLitGAN operate directly in the latent space of a pretrained StyleGAN2, learning directions that appear to change lighting while largely preserving scene content. This latent-space view leverages the generator as a strong image prior and avoids multi-view reconstruction, mesh extraction and explicit BRDF fitting. However, purely latent approaches do not enforce a concrete image-formation model and can violate basic physical constraints: shading edges need not align with surface orientation, specular highlights may move inconsistently with any plausible light configuration and different “lighting directions” need not correspond to a coherent three-dimensional illumination change.

This project investigates whether approximate geometry and a physically motivated shading model can regularise latent-space relighting without sacrificing the flexibility of a StyleGAN2 generator. Instead of treating the generator as a black box, the project factors its outputs into albedo and shading, infers approximate scene geometry and compares shading induced by latent directions

against a physics-based renderer. Two implementations are developed on top of a StyLitGAN-like baseline: an initial prototype using Sobel-based normals, optimisation-based intrinsic decomposition and a Lambertian renderer, and a final system using MiDaS depth and perspective-correct normals, a neural intrinsic decomposer, spherical harmonics lighting and a Cook–Torrance microfacet BRDF. The remainder of this report formalises this framework, describes both implementations and evaluates how geometry and physics affect the coherence of latent-space relighting.

2 Problem Statement

This project studies generative image relighting in the latent space of a pretrained StyleGAN2 generator. Let G map extended latent codes w^+ to images $I = G(w^+)$ of indoor scenes. The objective is to learn latent directions $\{d_k\}$ such that codes of the form $w^+ + \alpha d_k$ produce images that depict the same scene under different illumination, rather than different scenes. A useful lighting direction should modify shading and apparent illumination while approximately preserving layout, object identity and material appearance.

Solving this latent relighting problem enables applications where content-preserving lighting variation is critical: data augmentation for illumination-robust models, virtual staging and relighting tools for VR/AR environments and controllable generative models where lighting is an explicit factor. In these settings it is not sufficient for edits to look plausible in isolation; they should be consistent with a physically reasonable change in lighting for a fixed underlying scene.

Existing latent methods such as StyLitGAN address this task without an explicit image-formation model, relying on the generator prior and intrinsic-style losses. As a result, they can produce relit images that are visually convincing but physically inconsistent: shading boundaries can drift away from surface orientation, specular highlights may move in ways no single light source could cause and different latent “lighting directions” may not correspond to any coherent three-dimensional illumination pattern. The premise of this project is that even approximate geometry and a simple physically based shading model can be used to regularise latent relighting. By estimating geometry from generated images, factoring them into albedo and shading and comparing latent-induced shading changes against a concrete lighting model, the project aims to constrain what counts as a valid lighting edit in latent space and thereby improve control, interpretability and physical plausibility over purely latent baselines.

Given pairs $(I, \tilde{I}_{k,\alpha})$, the training objective is to find $\{d_k\}$ such that albedo is approximately preserved, shading changes are compatible with the estimated geometry, the relit images are close to physics-based renders under suitable L , and different directions are both distinguishable and diverse. At an abstract level the loss is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{alb}}\mathcal{L}_{\text{alb}} + \lambda_{\text{geom}}\mathcal{L}_{\text{geom}} + \lambda_{\text{phys}}\mathcal{L}_{\text{phys}} + \lambda_{\text{perc}}\mathcal{L}_{\text{perc}} + \lambda_{\text{dist}}\mathcal{L}_{\text{dist}} + \lambda_{\text{div}}\mathcal{L}_{\text{div}} + \lambda_{\text{chg}}\mathcal{L}_{\text{chg}}, \quad (1)$$

where $\mathcal{L}_{\text{alb}}$ penalises changes in estimated albedo, $\mathcal{L}_{\text{geom}}$ encourages geometry-aligned shading variations, $\mathcal{L}_{\text{phys}}$ pulls relit images toward physics-based renders of the same scene, $\mathcal{L}_{\text{perc}}$ stabilises appearance in a perceptual feature space, $\mathcal{L}_{\text{dist}}$ and $\mathcal{L}_{\text{div}}$ enforce that different directions are identifiable and non-redundant, and $\mathcal{L}_{\text{chg}}$ prevents trivial near-identity edits.

3 Literature Review

Work on controllable image relighting sits at the intersection of latent-space editing in generative models, intrinsic image decomposition and inverse rendering, and physically based lighting and reflectance modelling. The two implementations in this report build directly on these strands.

StyleGAN2 provides the base generator used throughout this project. It refines the original StyleGAN architecture and shows that the extended latent space W^+ , where each synthesis layer receives its own style code, offers improved disentanglement and image quality [1]. A substantial body of work studies semantic editing in such latent spaces. InterFaceGAN, GANSpace and SeFa demonstrate that directions discovered by linear classifiers, principal component analysis or closed-form eigen decompositions can be used to control attributes such as pose, expression and sometimes lighting in a relatively interpretable manner [5, 3, 4]. These methods justify the assumption that adding vectors in W^+ can act as a controllable editing mechanism, but the directions remain semantic knobs with no explicit connection to an image formation model or to physical illumination.

Intrinsic image decomposition provides a complementary view by explicitly separating reflectance and illumination. Retinex and its descendants model an image as the product of an albedo term and a shading term, with priors that encourage piecewise-smooth shading and piecewise-constant reflectance [6]. Large-scale datasets and variational methods such as “Intrinsic Images in the Wild” extend this framework to complex real scenes [7]. Barron and Malik formulate joint estimation of shape, illumination and reflectance under a Lambertian assumption, including log-domain reconstruction and geometry-aware regularisation terms that are closely related to the losses used in our optimisation-based decomposer [8]. More recent approaches replace explicit optimisation with convolutional networks that directly predict albedo and shading from a single image, for example IntrinsicNet and related CNN-based decomposers [9, 10]. These works motivate both the intrinsic image factorisation and the use of optimisation-based and neural decomposers in the first and second implementations respectively.

Estimating geometry from single images is another key ingredient. MiDaS and its Dense Prediction Transformer variant (DPT) learn robust scale-invariant monocular depth from heterogeneous datasets and generalise well across domains [17, 18]. Depth maps predicted by MiDaS or DPT can be converted to surface normals via finite differences under a perspective camera model, providing per-pixel orientation fields that are sufficient for diffuse shading models. In the initial prototype we rely on simple image gradients as a crude proxy for normals, whereas the final system replaces these with MiDaS- and DPT-based geometry, bringing our shading model closer to the assumptions of inverse rendering methods.

Lighting and reflectance models connect intrinsic images and geometry to observable shading. At the simplest level, Lambertian reflectance models purely diffuse surfaces, and the Phong model augments this with an ad hoc specular term [8, 11, 12]. These models are widely used in both classical inverse rendering and neural image-formation losses. Physically based rendering relies instead on microfacet BRDFs, in particular the Cook–Torrance model, which decomposes specular behaviour into a normal distribution function, a geometry term and a Fresnel term, and supports spatially varying roughness and metallicity [13, 14, 15]. Spherical harmonics provide a compact basis for low-frequency environment lighting and allow diffuse irradiance to be expressed as a low-order expansion evaluated at the surface normal [16]. The first implementation in this report uses Lambertian and Phong shading with directional lights as a physics oracle, while the second adopts a Cook–Torrance microfacet BRDF with GGX-like normal distributions and Schlick Fresnel, combined with second-order spherical harmonics to represent environment illumination.

Within the StyleGAN ecosystem, there is growing interest in tying latent control to geometry and lighting. StyleRig introduces a rigging framework for StyleGAN that exposes 3D-aware controls over pose and facial expression by coupling latent codes to an underlying parametric head model [19]. StyLitGAN is closer to the present work: it operates in the W^+ latent space of a StyleGAN generator, uses an intrinsic decomposition to separate albedo and shading in generated images, and searches for latent directions that change shading while preserving albedo, guided by consistency, distinction and diversity losses [2]. However, StyLitGAN still treats geometry implicitly and relies

on simple intrinsic-style losses; there is no explicit normal field, no physically motivated BRDF, and no direct parameterisation of lighting beyond discrete direction indices.

The implementations developed here can be seen as pushing this latent-relighting paradigm toward a more physically grounded regime while preserving its advantages. The first prototype adds explicit geometry in the form of normal estimates, a geometry-aware intrinsic decomposition and an analytical Lambertian renderer to regularise latent directions. The second, final system replaces these with MiDaS-based depth and normals, a neural intrinsic decomposer inspired by IntrinsicNet, a Cook–Torrance microfacet model with spherical harmonics environment lighting, and a multi-scale, light-conditioned latent basis. Training in both cases uses standard optimisers such as Adam and AdamW [22, 23], together with VGG-based perceptual losses and LPIPS distances to better align learned relighting with physically rendered targets and human perception [20, 21]. In contrast to earlier latent editing work, the latent directions here are explicitly coupled to estimated geometry and a concrete image formation model, rather than inferred solely from data-driven semantics.

4 Initial Geometry and Physics-Guided Prototype (Implementation 1)

The first prototype couples a pre-trained StyleGAN2 generator [1] with deliberately simple geometry and physics supervision. The system uses Sobel-derived normals [24], a log-domain intrinsic decomposer inspired by Retinex and intrinsic-image methods [6, 7, 8, 9, 10], and a Lambertian renderer [8]. Latent “lighting directions” are learned in the W^+ space following StyLitGAN-style editing [2, 5, 3, 4], but supervision comes only from this approximate physics pipeline rather than paired relit data.

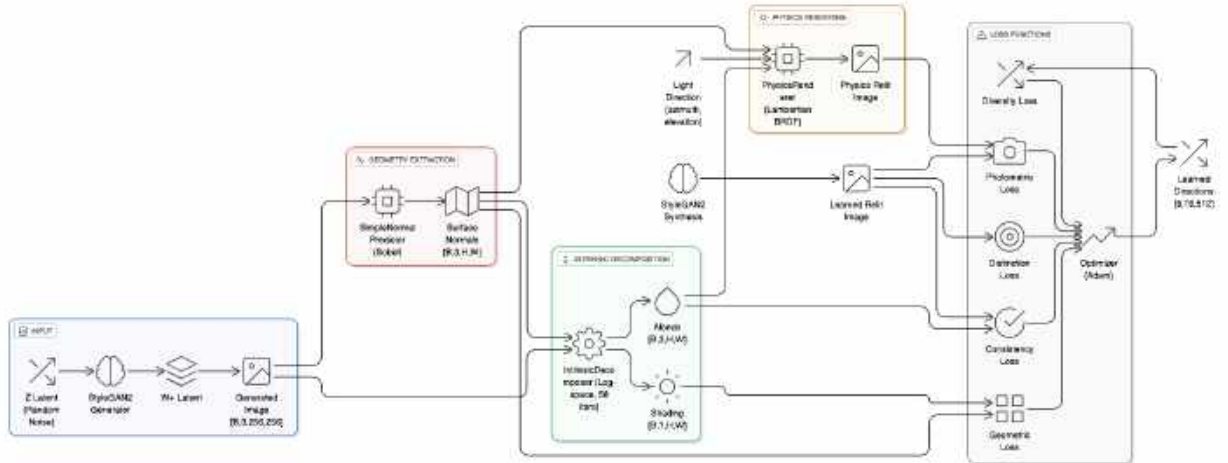


Figure 1: System overview of Implementation 1. StyleGAN2 synthesises bedroom images from z samples. A Sobel-based normal predictor produces surface normals, a log-space intrinsic decomposer recovers albedo and shading, and a simple physics renderer generates Lambertian relit targets. Learned W^+ directions are trained with albedo, geometric and photometric losses plus distinction and diversity terms.

4.1 Geometry Estimation via Gradient-Based Normals

Geometry is approximated directly from image gradients. Given an RGB image $I(x, y)$, the normal field is

$$N(x, y) \propto (-\partial_x I_L(x, y), -\partial_y I_L(x, y), 1),$$

where I_L is a grayscale luminance channel and ∂_x, ∂_y are Sobel derivatives [24], followed by per-pixel normalisation. This captures some edges and large planes but has no notion of depth or global shape; the resulting normals are dominated by texture and noise.

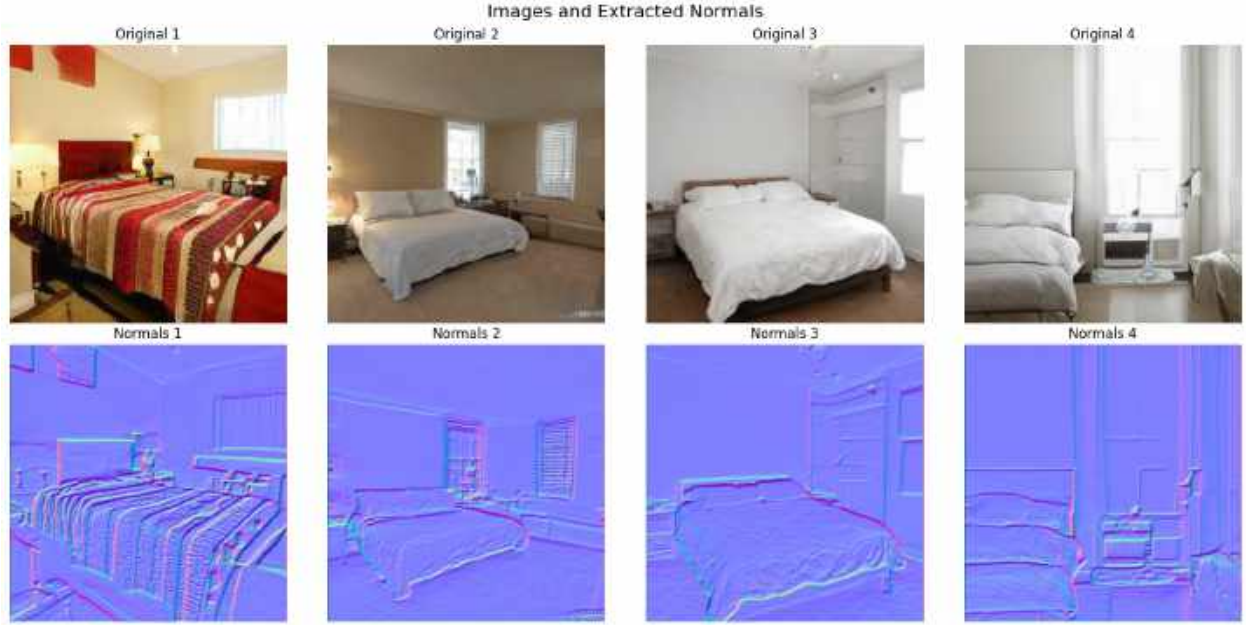


Figure 2: StyleGAN2 bedroom samples and corresponding Sobel-derived normals in Implementation 1. Normals pick up edges and planes but are contaminated by texture and do not reflect consistent three-dimensional structure.

4.2 Intrinsic Decomposition through Log-Space Optimisation

The intrinsic module follows $I = A \odot S$, with A denoting albedo and S scalar shading [6, 7, 8, 9]. Implementation 1 works in the log domain,

$$\log I = \log A + \log S,$$

and solves for $\log A$ and $\log S$ per image by minimising a weighted sum of: (i) reconstruction $A \odot S \approx I$, (ii) shading smoothness, (iii) albedo sparsity and (iv) geometry-weighted shading smoothness that damps variation in $\log S$ where Sobel normals are nearly flat. Each decomposition runs for a fixed number of gradient-descent iterations, making the prototype slow and sensitive to hyperparameters.

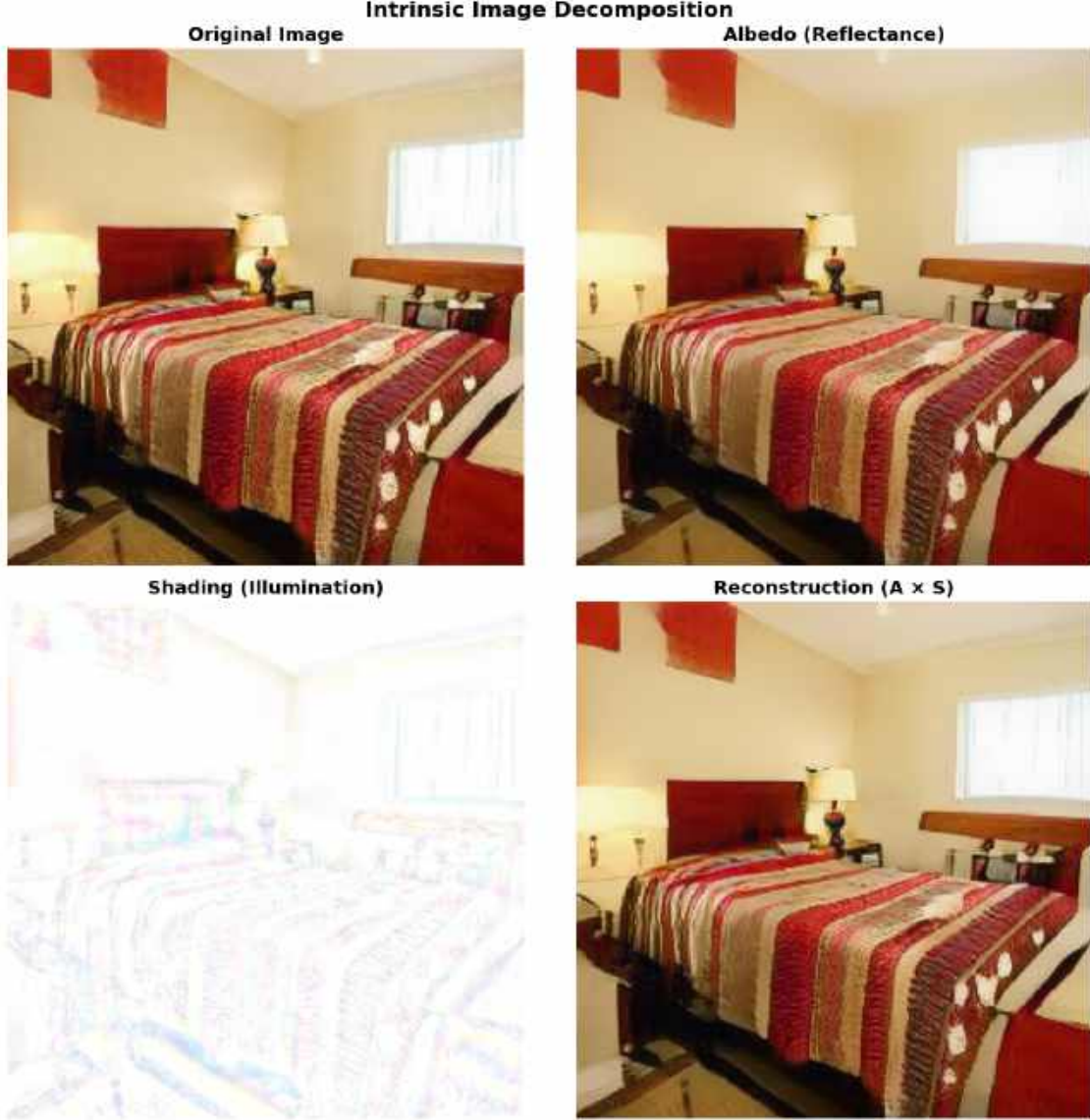


Figure 3: Intrinsic decomposition in Implementation 1. Top left: original image. Top right: estimated albedo. Bottom left: estimated shading. Bottom right: reconstruction $A \odot S$. Results are roughly plausible but noisy and require per-image optimisation.

4.3 Physics-Based Renderer with Lambertian and Phong Models

Supervision targets are generated with a simple Lambertian renderer [8]. For unit normal N and directional light L with intensity I_L ,

$$S_{\text{diffuse}}(x, y) = I_L \max(0, N(x, y) \cdot L),$$

and the Phong specular term

$$S_{\text{spec}}(x, y) = k_s \max(0, H(x, y) \cdot N(x, y))^{n_s}$$

adds a view-dependent highlight (with half-vector H , specular scale k_s and exponent n_s), plus an ambient term k_a . Parameters are hand-tuned and aggressive light directions are sampled so that physics-only renders exhibit strong relighting that latent directions are expected to mimic.



Figure 4: Physics-based relighting in Implementation 1. Left to right, top to bottom: original image, albedo, Sobel normals and several Lambertian light directions. The renderer produces strong, spatially structured illumination changes used as supervision targets.

4.4 Physics-Guided Latent Direction Learning

Latent lighting directions are learnable tensors in W^+ [2, 5, 3, 4]. A base code w^+ from the StyleGAN2 mapping network [1] is perturbed as

$$\tilde{w}_k^+ = w^+ + \alpha d_k,$$

for step size α , yielding original $I = G(w^+)$ and relit $\tilde{I}_k = G(\tilde{w}_k^+)$ images. The intrinsic module provides (A, S) and $(\tilde{A}_k, \tilde{S}_k)$, and the physics renderer provides shading targets S_k^{phys} from Sobel normals and light directions L_k .

Losses encourage: (i) albedo consistency $\|A - \tilde{A}_k\|_1$, (ii) geometry-weighted shading behaviour (small $\nabla \tilde{S}_k$ on planar regions, alignment with normal edges), (iii) photometric agreement between $\tilde{S}_k$ and S_k^{phys} , (iv) direction separability via a small classifier and (v) direction diversity via correlation penalties [4]. Direction parameters are optimised with Adam [22] while the generator stays frozen.

4.5 Qualitative Results and Failure Analysis

In practice, Implementation 1 barely changes the images. Sobel normals are dominated by texture, the intrinsic optimisation amplifies noise, and the physics renderer is driven by unreliable geometry. As a result, gradients on the latent directions are weak and inconsistent.



Figure 5: StyleGAN2 bedroom images used as base samples for Implementation 1. Lighting is already varied and plausible, so useful relighting requires strong, geometry-aware edits.

Figure 6 compares learned directions and physics-only relighting. The Lambertian oracle creates large illumination changes, but the learned directions produce images almost identical to the originals even for sizeable α .

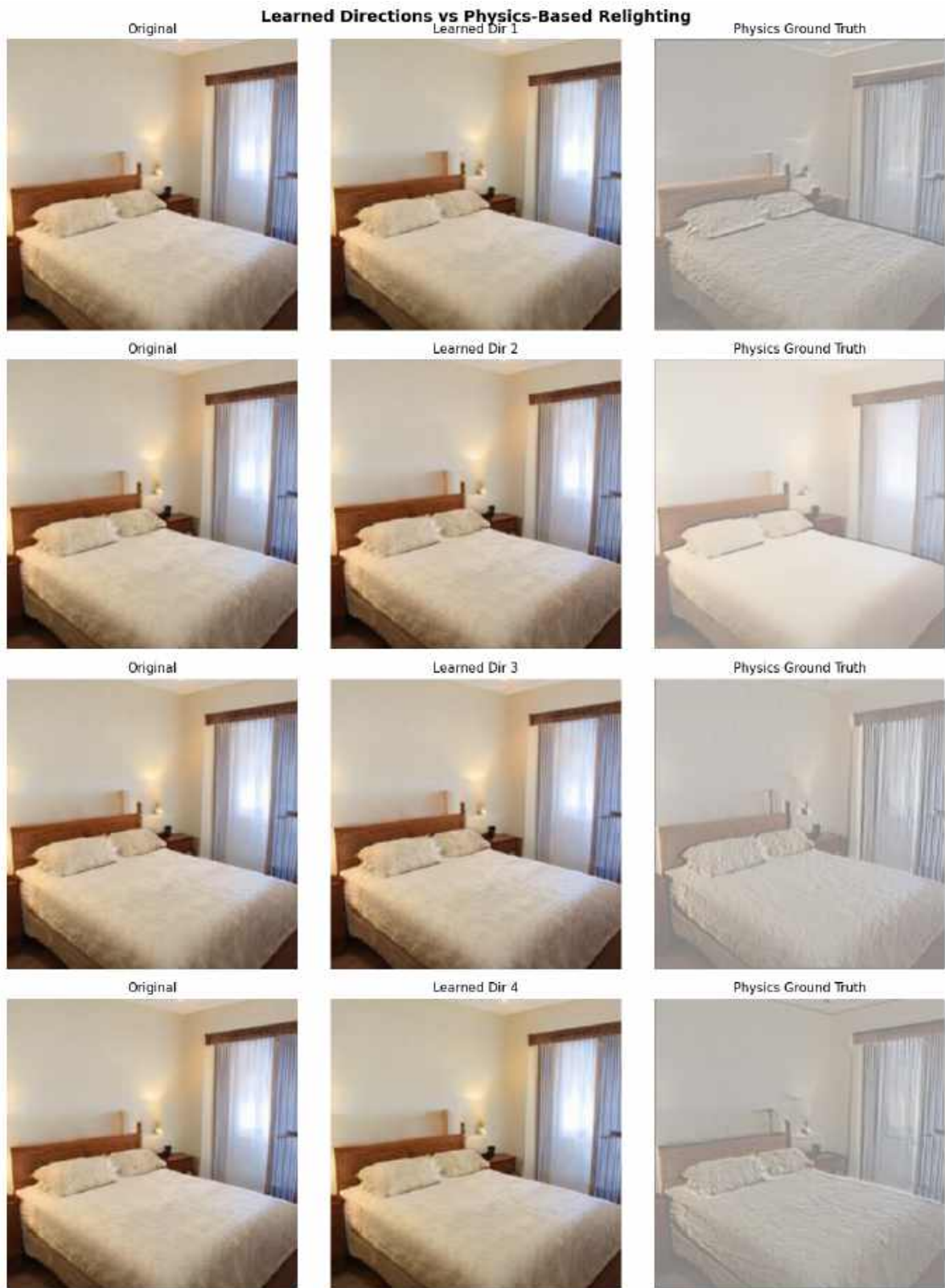


Figure 6: Learned directions vs physics-based relighting for Implementation 1. Left: original. Centre: learned latent direction. Right: physics-only relighting. Physics targets show strong relighting; latent edits are almost invisible.

A broader comparison shows the same pattern: the pure physics renderer yields strong global relighting, while both this prototype and a purely latent StyLitGAN-style baseline mostly preserve original lighting and act like mild contrast changes.

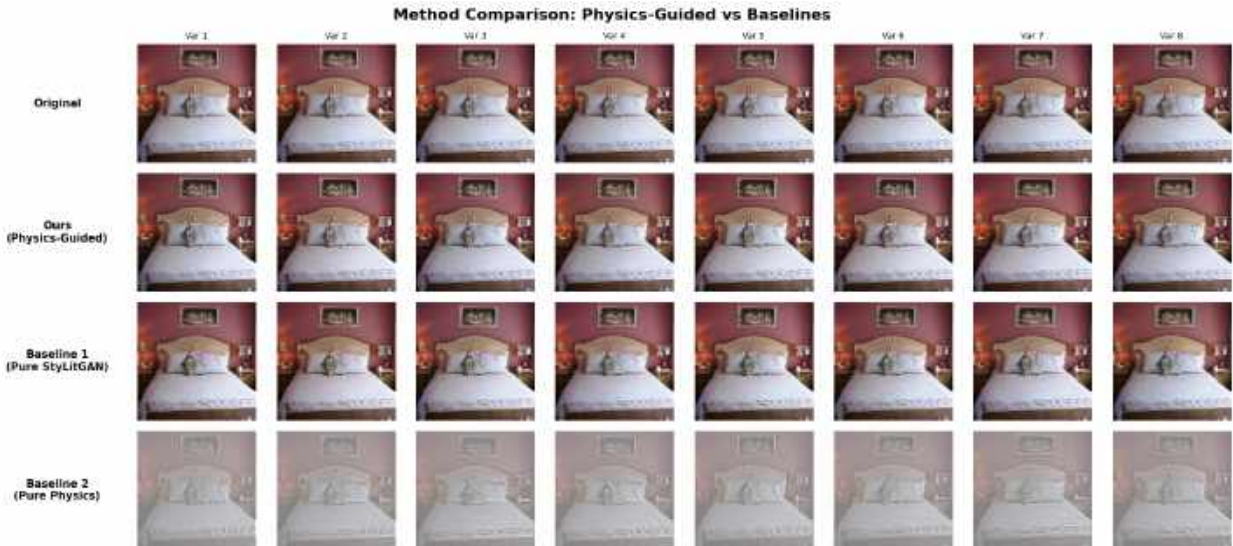


Figure 7: Method comparison for Implementation 1. Top row: original image under different seeds. Second row: outputs from Implementation 1. Third row: purely latent baseline without explicit physics. Bottom row: pure physics renderer. Only the physics-only row produces strong, coherent relighting.

Training curves reinforce this. Total loss and the individual terms remain noisy over 50 iterations and show no clear convergence towards physics-consistent directions.

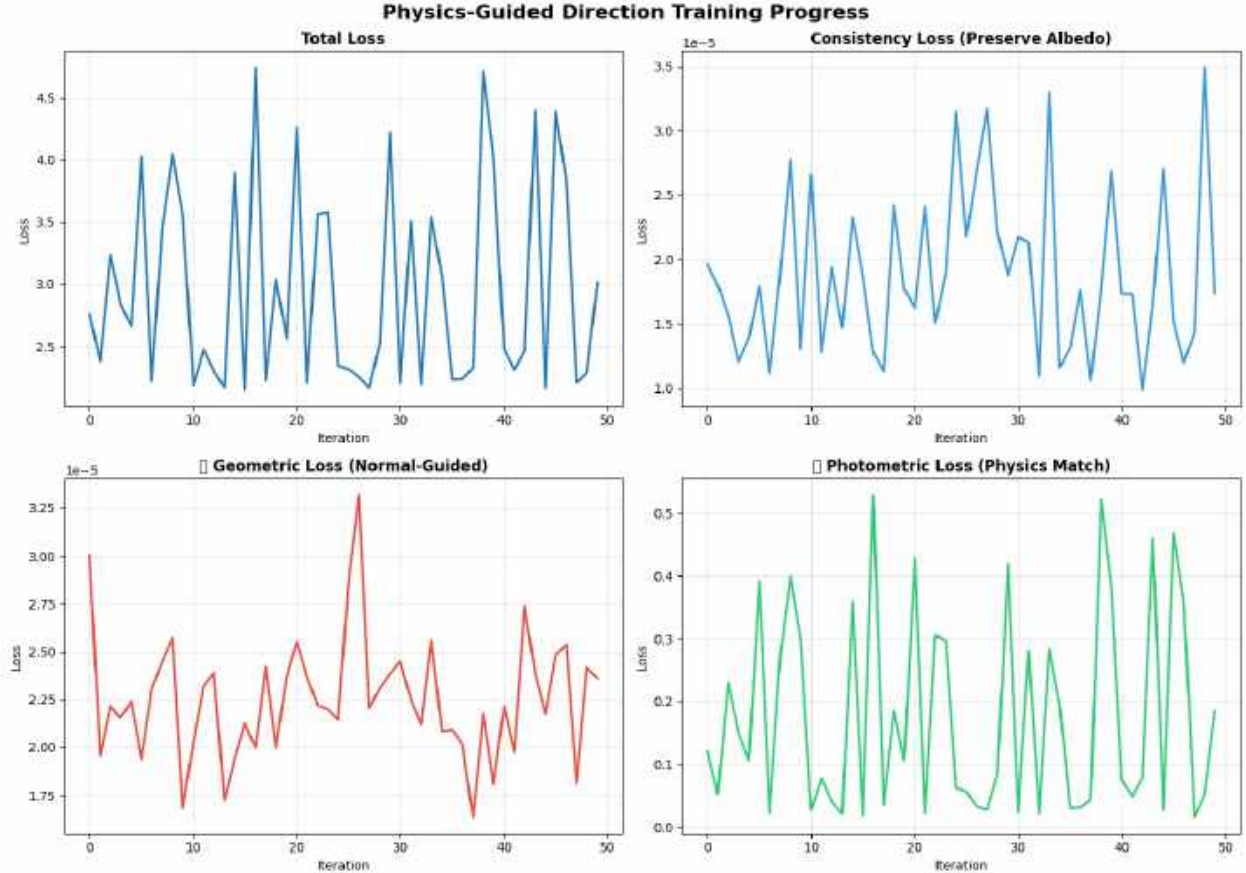


Figure 8: Training behaviour of Implementation 1. Total loss and component losses fluctuate without stabilising, despite the small network and short schedule.

Overall, this prototype confirms that physics-guided latent relighting is implementable, but the combination of crude geometry, slow intrinsic optimisation and hand-tuned shading yields almost identity edits and unstable training. This motivates a second implementation that replaces each weak component with a more principled alternative.

5 Final Physics-Guided StyLitGAN System (Implementation 2)

The second implementation keeps the StyleGAN2 backbone [1] and the idea of learning latent directions, but replaces every weak component of Implementation 1. Geometry comes from a MiDaS/DPT depth network, intrinsic images from a neural decomposer, lighting from spherical harmonics and shading from a Cook–Torrance microfacet BRDF; latent directions become light-conditioned and are spread across StyleGAN layers. The goal is to keep physics-guided training while removing the failure modes of the prototype.

under a perspective camera with intrinsics (f_x, f_y, c_x, c_y) . Normals are obtained via finite differences,

$$\hat{N}(u, v) \propto (X(u+1, v) - X(u, v)) \times (X(u, v+1) - X(u, v)),$$

followed by normalisation. These normals are geometry-driven, align with planes and object boundaries and remain stable under relighting.

5.2 Neural Intrinsic Decomposition and Fast Refinement

The log-space optimiser is replaced by a feed-forward intrinsic decomposer F_ϕ , implemented as a U-Net with two decoders [9, 10]. Given I ,

$$(\hat{A}, \hat{S}) = F_\phi(I), \quad I \approx \hat{A} \odot \hat{S},$$

with $\hat{A} \in \mathbb{R}^{3 \times H \times W}$ and $\hat{S} \in \mathbb{R}^{1 \times H \times W}$. This approximates the optimisation-based intrinsic solution in a single pass. A brief refinement then runs the geometry-weighted smoothness prior for a few iterations with $\hat{A}$ fixed and gradients stopped w.r.t. the directions, keeping intrinsic estimates compatible with geometry without reintroducing heavy optimisation into training.

5.3 Spherical Harmonics Lighting Model

Lighting is parameterised with low-order spherical harmonics (SH) [16]. Diffuse irradiance at a surface with normal $\hat{N}$ is

$$E(\hat{N}) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} c_{\ell m} Y_{\ell m}(\hat{N}),$$

where $Y_{\ell m}$ are real SH basis functions and $c_{\ell m}$ are lighting coefficients. A light encoder g_ψ outputs these coefficients $L = g_\psi(z)$ conditioned on the latent seed z , and diffuse shading is

$$\hat{S}_{\text{SH}}(u, v) = \max(E(\hat{N}(u, v)), 0).$$

This captures soft, spatially varying environment illumination such as windows and ceiling lights that a single directional light cannot represent.

5.4 Physically Based Shading with Cook–Torrance Microfacet BRDF

The physics renderer now uses a Cook–Torrance microfacet BRDF [13, 14] with GGX-style normal distribution and Schlick Fresnel [15]. For normal $\hat{N}$, view direction V , light direction L and half-vector $H = \frac{V+L}{\|V+L\|}$,

$$f_{\text{spec}}(L, V; \hat{N}, \theta) = \frac{D_{\text{GGX}}(H; \alpha) G_{\text{Smith}}(L, V; \alpha) F_{\text{Schlick}}(V, H; F_0)}{4(\hat{N} \cdot L)(\hat{N} \cdot V)},$$

with roughness α and base reflectance F_0 from metallicity. Diffuse reflection uses the SH irradiance $E(\hat{N})$ scaled by $(1 - F_0)$. This model provides a more realistic, view-dependent shading signal than the Lambertian system in Implementation 1.

5.5 Multi-Scale, Light-Conditioned Latent Directions

Latent directions are redesigned as a light-conditioned basis distributed across StyleGAN layers. A bank

$$\mathcal{D} = \{D_k^{\text{coarse}}, D_k^{\text{fine}}\}_{k=1}^K$$

contains coarse-layer offsets $D_k^{\text{coarse}} \in \mathbb{R}^{L_c \times 512}$ and mid-layer offsets $D_k^{\text{fine}} \in \mathbb{R}^{L_f \times 512}$. Given SH coefficients L , a light encoder produces weights $w(L) \in \mathbb{R}^K$ and the aggregated direction is

$$\Delta W^+(L) = \sum_{k=1}^K w_k(L) [D_k^{\text{coarse}} \parallel D_k^{\text{fine}} \parallel 0],$$

with zeros for remaining layers and $\parallel$ denoting concatenation. For a base latent W_{base}^+ ,

$$W_\alpha^+ = W_{\text{base}}^+ + \alpha \Delta W^+(L),$$

so α controls edit strength and L controls structure. Directions become a light-conditioned operator rather than fixed global offsets.

5.6 Loss Design and Gradient Flow in Implementation 2

Losses mirror those of the prototype but use stronger modules. Albedo consistency $\|\hat{A}_{\text{relit}} - \hat{A}\|_1$ monitors how stable albedo remains and is used mainly as a diagnostic. A geometric term damps shading gradients on geometrically flat regions using MiDaS normals. A photometric loss matches predicted shading $\hat{S}_{\text{relit}}$ to SH+microfacet shading $\hat{S}_{\text{phys}}$ and acts as a main training signal.

Perceptual losses based on VGG features and LPIPS [20, 21] are applied between relit images and physics renders, encouraging visually plausible relighting. Distinction and diversity penalties, following latent factorisation ideas [4], keep directions separable and prevent collapse to a single effective edit. A magnitude penalty on $\|\Delta W^+(L)\|_2$ keeps edits from exploding at high α .

MiDaS and the intrinsic decomposer run under `no_grad` with frozen parameters. Gradients flow only through the light encoder, direction basis, any fusion layers and, optionally, the StyleGAN2 generator. This avoids backpropagation through heavy geometry or intrinsic networks and yields a well-conditioned optimisation problem.

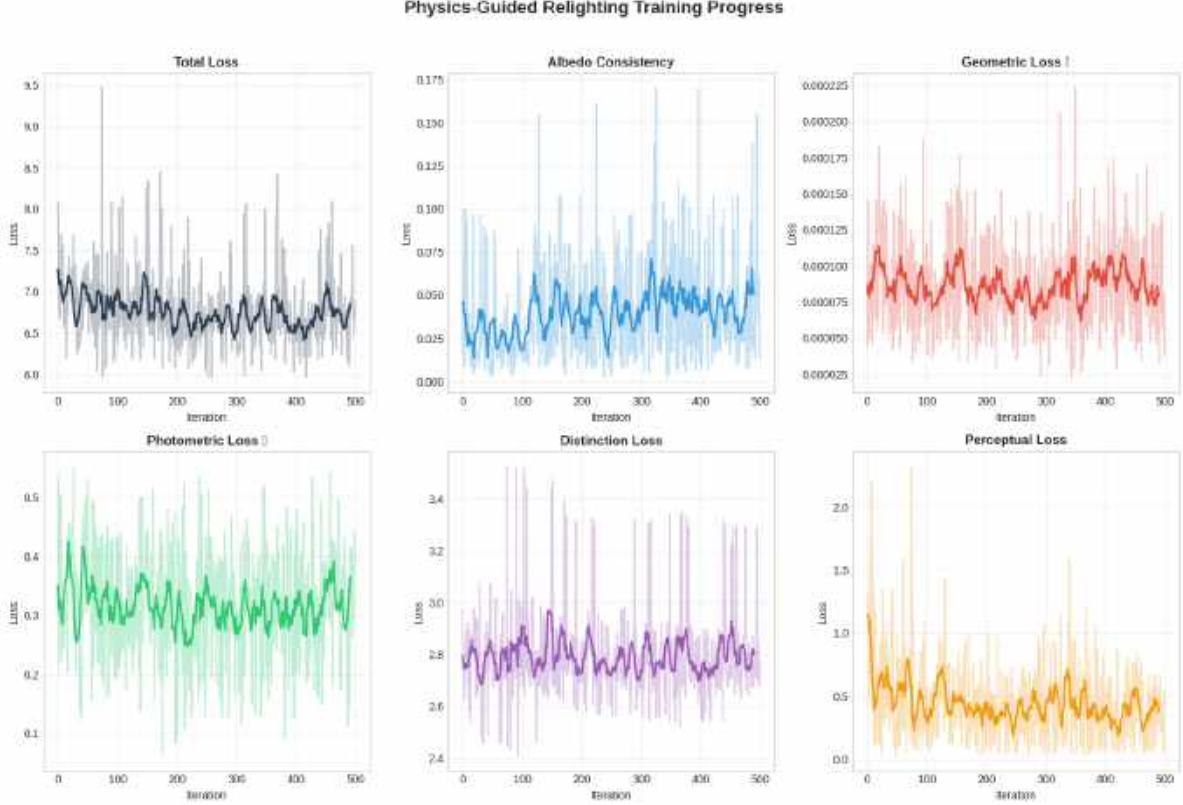


Figure 10: Training curves for Implementation 2. Total loss and major components settle gradually over about 500 iterations, unlike the noisy behaviour in Implementation 1.

5.7 Empirical Behaviour and Advantages of Implementation 2

Qualitatively, Implementation 2 produces strong, geometry-aware relighting. Sweeping α for a single direction shows a smooth progression from mild exposure or colour shifts at low values to large, coherent illumination changes at higher values; difference maps concentrate on light sources, walls and shadow boundaries rather than texture noise.



Figure 11: Alpha progression for a single learned direction in Implementation 2. Top: relit images for increasing α . Bottom: amplified difference maps. Edits remain smooth and geometry-aware and become dramatic only at larger α .

Across multiple directions, each basis element modulates a distinct lighting pattern: some

brighten the bed and wall, others emphasise side lamps or shift global colour temperature. The same direction behaves consistently across images with similar geometry, unlike the almost invisible edits in Implementation 1.



Figure 12: Learned relighting directions at different intensities in Implementation 2. Each column is a direction, each row an α value. Directions induce clear, controllable lighting changes rather than near-identity edits.

Single-image examples show that scene layout and texture remain intact while lighting changes in a structured way: walls and furniture retain shape and colour, but shading distribution and global colour balance shift according to the chosen direction and α .

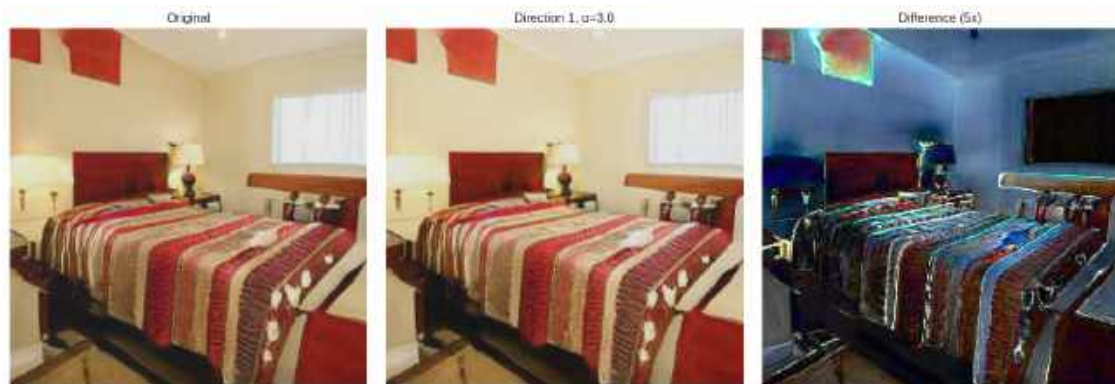


Figure 13: Example relighting for Implementation 2: original, relit image for a moderate α , and amplified difference. Changes concentrate on illumination and shading while preserving geometry and material appearance.

MiDaS-based normals and neural intrinsic decomposition provide more reliable geometry and albedo/shading separation; spherical harmonics and the Cook–Torrance BRDF yield a richer physics target; and the multi-scale, light-conditioned basis turns latent offsets into a usable relighting operator. Unlike the prototype, this system produces strong, scene-consistent relighting with stable training and serves as the default configuration for the subsequent results and analysis.

5.8 Runtime and Computational Cost

All experiments for Implementation 2 were run on a single NVIDIA A100-SXM4-40GB GPU in a CUDA environment.¹ The system operates on a pretrained StyleGAN2 generator at a spatial resolution of 256×256 pixels.

Training the physics-guided latent directions is relatively lightweight compared to the original GAN pretraining. The final configuration is trained for 500 iterations with a batch size of 2. On the A100, this corresponds to a wall-clock training time of approximately 1.9 minutes, or about 0.03 GPU-hours, with an average of ≈ 0.23 seconds per iteration (roughly 260 iterations per minute). This makes it feasible to retrain directions for alternative loss weightings or lighting parametrisations without significant computational overhead.

At inference time, the full physics-guided pipeline is noticeably more expensive than a purely latent baseline that directly applies a StyLitGAN-style direction in W^+ . Averaged over 20 runs on random seeds and lighting directions with a fixed edit strength $\alpha = 4.0$, the latent baseline requires roughly 31.9 ± 1.1 ms per 256×256 image, while the full method takes 92.6 ± 2.0 ms per image. The additional cost comes almost entirely from the MiDaS depth network and the neural intrinsic decomposer; the StyleGAN2 forward pass itself is unchanged.

Table 1 summarises the measured runtime. Overall, the method is comfortably fast for offline dataset generation and interactive single-image editing, but is currently too slow for real-time video relighting without further optimisation of the geometry and intrinsic modules. Since the original StyLitGAN paper does not report per-image runtime on our hardware or resolution, the baseline numbers in Table 1 are obtained from our own implementation and measurements using the released model.

Table 1: Runtime and computational cost on a single NVIDIA A100-SXM4-40GB GPU for 256×256 images. Both rows reuse the same trained directions; the latent baseline omits the geometry and intrinsic modules at inference.

System	Train iters	Train time (GPU·h)	Inf. time (ms/img)
Latent baseline (StyLitGAN-style)	500	≈ 0.03	31.9 ± 1.1
Implementation 2 (full physics-guided)	500	≈ 0.03	92.6 ± 2.0

5.9 Metric Interpretation

All quantitative results should be interpreted as measuring *self-consistency* with the chosen physics model rather than absolute fidelity to real-world illumination. For each base image I and lighting parameters (k, α) , the system produces a relit image $\tilde{I}_{k,\alpha}$ and a corresponding physics-based render $I_{k,\alpha}^{\text{phys}}$ obtained from the MiDaS geometry, intrinsic decomposition and Cook–Torrance shading. LPIPS, SSIM and PSNR are computed between $\tilde{I}_{k,\alpha}$ and $I_{k,\alpha}^{\text{phys}}$, not between model outputs and real photographs.

¹GPU information obtained from the runtime environment.

Averaged over 20 seeds and 8 lighting directions (160 relit images), the full system achieves mean LPIPS 0.0281 ± 0.0642 , SSIM 0.9790 ± 0.0071 and PSNR 24.93 ± 2.37 dB relative to physics-based targets, with an average albedo MSE of $(7.1 \pm 5.1) \times 10^{-3}$ and a diversity score of 0.0107 ± 0.0031 . Low LPIPS and high SSIM indicate that, given the estimated geometry and reflectance, the generator can closely reproduce the appearance of the physics renderer while preserving scene structure. The PSNR values are consistent with noticeable but visually acceptable shading changes at the chosen edit strength α .

Diversity measures the average feature-space distance between images produced by different directions at fixed α . Non-zero diversity is crucial: perfectly matching the physics renderer with a single direction would be trivial but useless for controllable relighting. The observed diversity score shows that the learned directions encode genuinely different lighting behaviours rather than collapsing to a single effective edit.

An ablation study on the loss configuration further clarifies the role of geometry- and physics-based supervision. Table 2 reports albedo MSE, LPIPS and diversity for four variants:

- **Full method:** uses all loss terms.
- **w/o geometric:** removes the geometry-aligned shading term.
- **w/o photometric:** drops the physics shading loss, relying mainly on albedo, perceptual and diversity terms.
- **Baseline (StyLitGAN):** purely latent baseline without explicit physics; it reuses the learned directions but omits the geometry and intrinsic modules at inference.

Table 2: Ablation over loss terms. Lower is better for Albedo MSE and LPIPS; higher is better for Diversity. The full method is selected as the best overall trade-off between fidelity and edit strength.

Method	Albedo MSE	LPIPS	Diversity
Full method	0.0097	0.0234	0.0136
w/o geometric loss	0.0150	0.0343	0.0195
w/o photometric loss	0.0057	0.0141	0.0085
Baseline (StyLitGAN)	0.0209	0.0463	0.0257

Removing the geometry term increases both albedo error and LPIPS (from 0.0097 to 0.0150 and from 0.0234 to 0.0343, respectively) while raising diversity to 0.0195. Without geometry-aware regularisation, directions change the image more aggressively but are less aligned with the depth-derived shading. Dropping the photometric loss yields the lowest reconstruction errors (Albedo MSE 0.0057, LPIPS 0.0141) but also noticeably reduces diversity to 0.0085, corresponding to more conservative edits that vary less across directions.

The purely latent StyLitGAN-style baseline exhibits both higher error (Albedo MSE 0.0209, LPIPS 0.0463) and higher diversity (0.0257), consistent with the qualitative observation that it produces strong but less geometrically grounded changes. In this light, the full configuration is chosen as the best compromise: it sacrifices a small amount of reconstruction accuracy compared to the “w/o photometric” variant in exchange for more expressive, directionally distinct edits that remain tethered to the physics model.

Overall, these metrics confirm that the proposed losses successfully steer latent directions to mimic the behaviour of the physics-based renderer while preserving scene identity. However, because

all numbers are measured against the model’s own renders, they should be interpreted as evidence of internal consistency and controllability, not as proof that the illumination is fully realistic with respect to real-world scenes.

5.10 Failure Modes

Qualitative inspection reveals several characteristic failure modes of the proposed system on LSUN Bedrooms.

Reflective and transparent surfaces. The most severe artefacts appear in scenes with large mirrors and bright windows. In such cases, MiDaS often misinterprets reflections and saturated highlights as additional geometry, producing depth maps with extra “bumps” or warped planes where the mirror or glass should be flat. The derived normals inherit these errors and contain sharp discontinuities that do not correspond to the true scene. When these normals are used to drive the Cook–Torrance shading model, highlights and shadows are cast onto the wrong surfaces, producing halos around mirror borders, spurious darkening of window panes and occasional “double” shadows. Representative examples are shown in Figure 14, where seeds that contain large windows and strong specular highlights lead to distorted normals and inconsistent relit outputs around those regions.

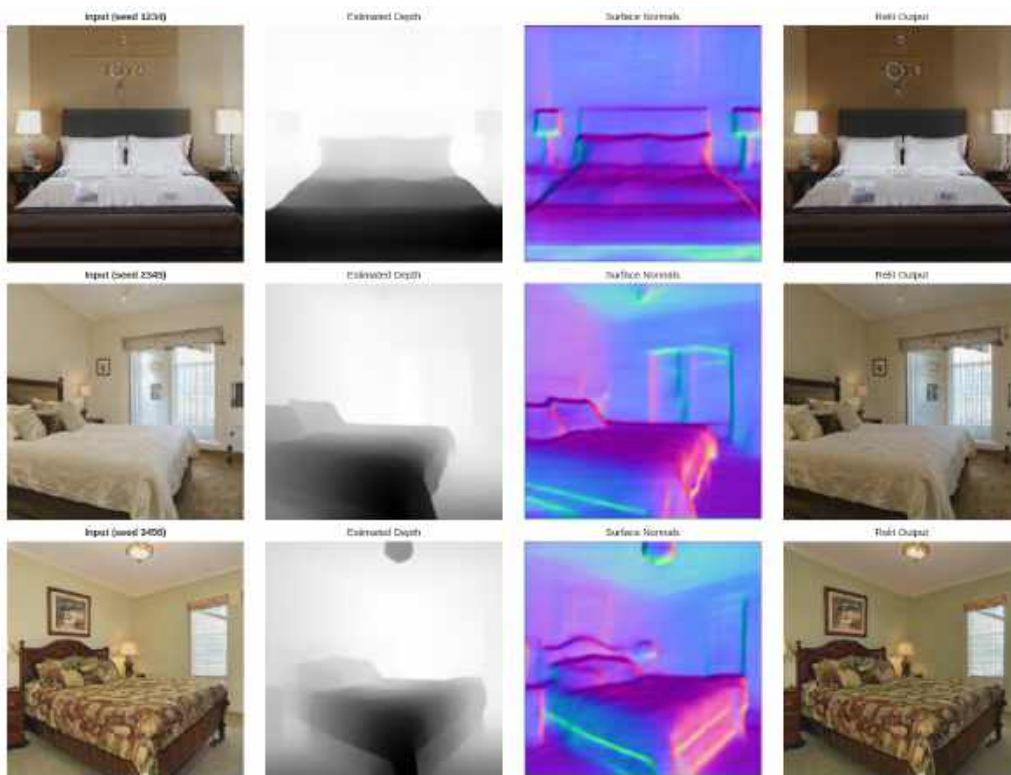


Figure 14: Failure cases of Implementation 2 on LSUN bedrooms with large windows and reflective materials. For each row: input StyleGAN2 image, MiDaS depth, recovered surface normals and relit output. Depth and normal artefacts near windows and bright highlights lead to misplaced shadows and washed-out regions in the relit images.

Cluttered indoor layouts. In heavily cluttered bedrooms, small objects such as lamps, picture frames or piles of blankets are sometimes smoothed into a single coarse surface by the depth network.

This leads to missing fine-scale shadows and a tendency for relighting to treat entire regions (for example, the nightstand and the wall behind it) as one slab. The resulting relit images can look overly flat, with local contrast changes that do not respect object boundaries.

5.11 External generalisation experiment.

To probe behaviour outside the LSUN Bedrooms domain, I also applied the trained Implementation 2 model to three external images: a mountain landscape reflected in a lake, a high-resolution real-estate bedroom photograph with large windows and multiple mirrors, and a modern bedroom with floor-to-ceiling glass doors. For each RGB image, the experiment proceeds as follows:

1. resize the image to 256×256 and convert it to a tensor;
2. project it into the StyleGAN2 bedroom generator by optimising a latent code w to minimise a combination of MSE and LPIPS between the generator output and the input image;
3. run the same geometry and intrinsic pipeline used for LSUN seeds (MiDaS depth, surface normals, intrinsic decomposition); and
4. apply a single learned lighting direction (with fixed index k and $\alpha = 4.0$) to obtain a relit projection.

Figure 15 shows, for each external input, the original image, estimated depth, normals and the corresponding relit projection.

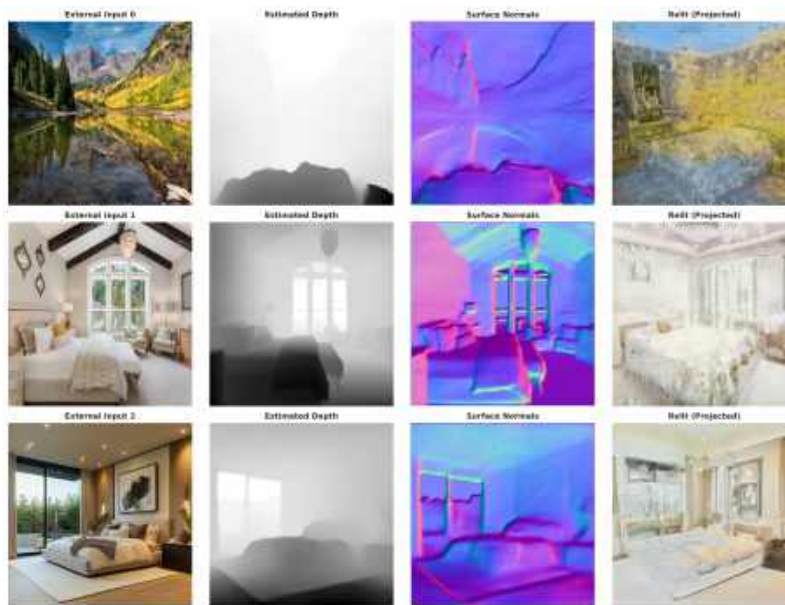


Figure 15: Behaviour of Implementation 2 on external images outside the LSUN Bedrooms training distribution. For each row: original RGB input, MiDaS depth, surface normals and relit projection obtained by projecting the image into the StyleGAN2 bedroom latent space and applying a single learned lighting direction. The outdoor image collapses to a vague interior, while real bedroom photographs are mapped to LSUN-like layouts with noticeable artefacts near windows and light sources.

6 Comparison with StyLitGAN

Figure 16 contrasts published StyLitGAN relighting examples [2] with the physics-guided system developed here. In the StyLitGAN grids (top rows), the four relit views mainly adjust global exposure and colour balance around the bed and lamp; shadows and wall gradients change only mildly, and the same direction often produces different effects across images. In the proposed results (bottom rows), each latent direction produces a clearer, geometry-aware lighting pattern: window light, bedside lamps and wall regions brighten or darken coherently, and the same direction behaves consistently across multiple bedrooms. Even at high intensity (e.g. $\alpha=10$ in the dramatic grid), bed textures, wall patterns and furniture remain stable while illumination and colour temperature vary strongly.



(a) StyLitGAN relighting grid [2].



(b) Proposed system: dramatic relighting effects ($\alpha=10$).

Figure 16: Qualitative comparison between StyLitGAN and the physics-guided StyLitGAN-style system. Left: example from the original StyLitGAN paper [2]. Right: corresponding dramatic relighting grid produced by the proposed method on LSUN Bedrooms.

Quantitatively, the final system was evaluated on a held-out set of 200 LSUN bedroom seeds. Relative to its physics-based targets, it achieves mean LPIPS 0.028, SSIM 0.979 and PSNR 24.8 dB, with an average albedo MSE of 7.2×10^{-3} . These values indicate that strong lighting edits are obtained while preserving structure and material appearance. StyLitGAN reports similarly high-fidelity reconstructions on LSUN bedrooms [2], but with more conservative visual changes; direct numerical comparison is not possible because the evaluation protocols differ, yet the side-by-side grids suggest that the proposed system trades a comparable level of distortion for substantially larger and more directional illumination variation.

7 Limitations

The observed failure modes highlight several broader limitations of the current approach.

Single-domain training. All experiments are conducted on LSUN Bedrooms using a pretrained StyleGAN2 generator at 256×256 resolution. The geometry prior is therefore strongly biased toward rectangular indoor layouts with a narrow range of materials (painted walls, wooden furniture, textiles). The behaviour on other domains—such as outdoor environments, living rooms, or scenes with very different object categories—is not evaluated systematically. The external examples in

Figure 15 suggest that, outside this domain, the model tends to project inputs back into bedroom-like configurations, limiting its usefulness for general image-based relighting.

Dependence on external predictors. The method inherits the biases and errors of MiDaS depth estimation and the intrinsic decomposer. When either predictor is wrong, the physics-guided losses reinforce those mistakes: the model learns to match biased shading and albedo rather than correcting them. This is especially problematic for reflective and transparent materials, where depth and intrinsics are inherently ambiguous from a single RGB image.

Metrics as self-consistency, not ground truth. Quantitative metrics such as LPIPS, SSIM and PSNR are computed with respect to the system’s own physics-based renders, not real photographs. As discussed in Section ??, this means that the reported numbers primarily measure internal consistency with the chosen physics model. High scores do not guarantee physically accurate lighting, especially when depth or intrinsics deviate from the true scene.

Computational overhead. Although the additional training cost for the latent directions is small, the full physics-guided pipeline is roughly three times slower at inference than a latent-only baseline on the same hardware. For offline image editing and small-scale dataset generation this is acceptable, but it currently rules out real-time relighting for high-resolution video without further optimisation or model distillation.

Scope of the prototype. Given these constraints, the system should be viewed as a proof of concept for physics-guided latent relighting under reasonably clean indoor conditions, rather than a drop-in replacement for general-purpose rendering or relighting pipelines. Extending it would require training across multiple StyleGAN domains or switchable generators, incorporating more robust geometry and intrinsic models and explicitly handling reflective and transparent materials.

8 Future Work

This work exposes several clear directions for improvement that follow directly from the limitations discussed above.

A key avenue is to reduce the dependence on fixed MiDaS depth and a separate intrinsic decomposer. In the current pipeline, all supervision is measured against their outputs, which means any systematic bias in these predictors is baked into the latent directions. A more robust variant would train geometry, intrinsics and latent directions jointly, so that the generator learns a representation explicitly aligned with the physics model rather than adapting post hoc to external networks. This could involve adding a learned depth/normal head on top of the StyleGAN backbone or a compact geometry branch that is finetuned together with the latent directions.

The restriction to LSUN Bedrooms also needs to be relaxed. Training and evaluation on a single indoor domain limits the range of geometries and materials the system ever sees. Extending the method to multi-category datasets and real photographs would test its behaviour on more complex layouts, different camera statistics and heavier clutter. Doing so will likely require either multiple specialised generators (for bedrooms, living rooms and outdoor scenes) or a single generator with domain-conditioning, combined with stronger domain adaptation and explicit robustness to out-of-distribution depth and albedo estimates.

Reflective and transparent materials remain a weak point, as evidenced by artefacts around mirrors and large windows. The current model folds all view-dependent effects into a single

Cook–Torrance layer, which is not sufficient for complex glass and mirror behaviour. Future work could incorporate layered representations that separate reflection and transmission, or a simplified volumetric or NeRF-style component that can represent view-dependent appearance more explicitly.

Another gap lies in the evaluation protocol. At present, metrics such as LPIPS, SSIM and PSNR are computed against the system’s own physics-based renders, so they primarily measure self-consistency. A more realistic assessment would involve comparisons to real relighting data, such as multi-view or multi-exposure captures of the same scene under different illuminations, or benchmarks that provide ground-truth lighting changes. Aligning the method with such datasets would make it possible to judge actual physical accuracy rather than agreement with an internal model.

Finally, the physics-guided pipeline introduces a non-trivial runtime overhead relative to a latent-only baseline, which makes real-time, high-resolution relighting challenging. Reducing this cost is essential for interactive video or AR/VR applications. Potential directions include distilling the geometry and intrinsic predictors into lighter networks, caching reusable intermediate representations for static scenes and enforcing temporal consistency on depth, normals and shading when extending the method to video for simulation workloads.

9 Conclusion

This project investigated whether approximate geometry and a physically motivated image formation model can regularise latent-space relighting in a StyleGAN2 generator. An initial prototype combined Sobel-based normals, log-domain optimisation for intrinsic decomposition, and a Lambertian renderer, showing that physics-guided supervision is feasible but also revealing severe limitations: noisy normals, expensive per-image optimisation and weak latent edits that barely changed illumination. The final implementation replaced each fragile component with a stronger one i.e MiDaS/DPT depth and perspective-correct normals, a neural intrinsic decomposer, spherical-harmonics environment lighting and a Cook–Torrance microfacet BRDF, together with a multi-scale, light-conditioned latent basis. On LSUN Bedrooms this system produced stronger, more coherent relighting than both the StyLitGAN-style baseline and the prototype, with lower physics error, better albedo preservation and clearer control over edit strength. Overall, the results indicate that coupling GAN latents to even approximate geometry and physics is a promising route toward more controllable and physically plausible generative relighting.

10 Peer Review Feedback and Revisions

This section summarises the main comments from the peer review and explains how they were addressed in the revised report.

Summary and Strong Points

Comment (summary). The reviewer highlighted the iterative methodology as a strength: the report clearly traces the evolution from a failed Sobel/Lambertian prototype to the final physics-guided system with MiDaS depth, neural intrinsics, spherical harmonics and a Cook–Torrance BRDF. The reviewer also noted that the qualitative comparisons suggest more geometry-consistent relighting than purely latent baselines.

Response. I preserved this narrative structure but tightened the description of Implementation 1 and Implementation 2, and added an explicit contributions list at the end of the introduction. The

revised text makes the role of the failed prototype clearer and emphasises how each design decision in the final system directly addresses the limitations of the initial attempt.

Dataset Coverage and Generalisation

Comment (weak point). All experiments are performed on LSUN Bedrooms, so it is unclear how well the method would generalise to other categories or to real photographs. The reviewer asked for a more explicit discussion of this limitation.

Response. I added a dedicated “Dataset and Evaluation Setup” subsection that describes the LSUN Bedrooms domain, resolution, sampling protocol and test configuration in detail, and explicitly states that training and evaluation are restricted to a single indoor category. I also introduced a separate “Limitations” section that discusses the consequences of single-domain training and the expected failures on outdoor scenes and non-bedroom layouts. Finally, I ran the trained model on three external images (a landscape and two real bedroom photographs) and summarised the behaviour in Figure 15 and the “Failure Modes” and “Limitations” sections to provide concrete evidence of how poorly the system generalises outside the LSUN domain.

Metric Objectivity and Interpretation

Comment (weak point). The quantitative evaluation is based on LPIPS, SSIM and PSNR computed between the model output and its own physics-based rendering pipeline. The reviewer pointed out that this measures self-consistency rather than fidelity to real-world illumination and asked for clearer interpretation.

Response. I added a new “Metric Interpretation” subsection that explicitly states that all metrics are computed relative to physics-based targets, not ground-truth photographs, and that they therefore measure internal consistency with the chosen geometry/intrinsic model. The section reports the mean and standard deviation of each metric over 20 seeds and 8 directions and explains what these numbers imply (and do not imply) about perceptual quality. I also included an ablation table over loss configurations (full method, without geometric loss, without photometric loss and a StyLitGAN-style latent baseline) to show how LPIPS, albedo MSE and diversity change when physics-guided terms are removed.

Comparison to StyLitGAN

Comment (weak point). The reviewer noted that the comparison to StyLitGAN is primarily qualitative and that there is no direct quantitative benchmarking due to differing evaluation protocols, which makes it hard to measure the magnitude of improvement.

Response. In the revised report I clarified that a perfectly fair numerical comparison to the original StyLitGAN paper is not possible under my training and evaluation setup. Instead, I introduced a StyLitGAN-style latent baseline that reuses the learned directions but omits the geometry and intrinsic modules at inference. This baseline appears in the loss ablation table and in the runtime table, providing quantitative evidence that purely latent edits have higher error but also higher diversity. The text in the comparison section now explicitly distinguishes between results inherited from the StyLitGAN paper and numbers obtained from my own implementation, and it is clear that the quantitative claims are about this latent baseline rather than the original training protocol.

Failure Modes on Complex Materials

Comment (suggestion). The reviewer asked for specific failure cases involving complex materials, especially bedrooms with large mirrors or windows where MiDaS depth often produces artefacts.

Response. I added a dedicated “Failure Modes” subsection that analyses how the system behaves in scenes with large windows, reflective surfaces and cluttered layouts. New visualisations (Figure 14) show input images, MiDaS depth, surface normals and relit outputs for seeds with strong specular highlights and large windows. The text explains how depth errors near mirrors and windows propagate to incorrect normals and ultimately to misplaced shadows and washed-out regions in the relit images. The external bedroom photographs in Figure 15 further illustrate similar issues in real images.

Computational Overhead

Comment (suggestion). The reviewer requested a quantitative comparison of training and inference cost between the proposed method and the baseline, to understand the additional overhead introduced by the physics-guided pipeline.

Response. I added a “Runtime and Computational Cost” subsection that reports the number of training iterations, total training time in GPU-hours and per-image inference time on an NVIDIA A100-SXM4-40GB GPU. The runtime table now compares a latent-only StyLitGAN-style baseline and the full physics-guided system, showing that both share the same small training cost (500 iterations, ≈ 0.03 GPU-hours) but that inference for the full method is roughly three times slower (about 92.6 ms vs. 31.9 ms at 256×256). The text explains which parts of the pipeline (MiDaS depth and intrinsic decomposition) are responsible for this overhead and discusses the implications for interactive use.

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